ROLL NO : 241801038

Ex 13: Implement your own Ping program.

Aim :

To implement a simple Ping program in computer networks to check whether a destination host is reachable and to measure the round-trip time (RTT) of packets.

Algorithm :

1. Start the program.
2. Read the destination hostname or IP address from the user.
3. Create a raw ICMP socket.
4. Construct an ICMP Echo Request packet containing type, code, checksum, identifier, sequence number, and data.
5. Calculate the checksum of the packet.
6. Send the ICMP Echo Request packet to the destination host.
7. Wait for an ICMP Echo Reply from the destination.
8. Calculate the round-trip time (RTT) between sending the request and receiving the reply.
9. Display the destination IP address, packet size, sequence number, and RTT.
10. Repeat the process for a specified number of packets.
11. Display the final result and terminate the program.

```
import socket
import struct
import time
import os

def checksum(data):
    total = 0
    length = len(data)

    for i in range(0, length - 1, 2):
        total += (data[i] << 8) + data[i + 1]
```

```
        if length % 2:
            total += data[-1] << 8

        total = (total >> 16) + (total & 0xffff)
        total += total >> 16

        return (~total) & 0xffff


    def ping(host):
        try:
            dest_addr = socket.gethostbyname(host)
        except socket.gaierror:
            print("Unable to resolve host")
            return

        icmp = socket.getprotobyname("icmp")

        try:
            sock = socket.socket(
                socket.AF_INET,
                socket.SOCK_RAW,
                icmp
            )
        except PermissionError:
            print("Permission denied. Run with administrator/root privileges.")
            return

        sock.settimeout(2)

        packet_id = os.getpid() & 0xFFFF
        sequence = 1

        header = struct.pack(
            "!BBHHH", 8, 0, 0, packet_id, sequence
        )

        data = b"Computer Networks Ping"
        packet = header + data

        chksum = checksum(packet)

        header = struct.pack(
            "!BBHHH", 8, 0, chksum, packet_id, sequence
        )

        packet = header + data
        start_time = time.time()

        try:
            sock.sendto(packet, (dest_addr, 1))
```

```
            sock.recvfrom(1024)

            end_time = time.time()
            rtt = (end_time - start_time) * 1000

            print(
                "Reply from", dest_addr,
                ": bytes =", len(packet),
                "time = %.2f ms" % rtt
            )

        except socket.timeout:
            print("Request timed out")

        finally:
            sock.close()


    host = input("Enter hostname or IP address: ")

    for i in range(4):
        ping(host)
        time.sleep(1)
```

```
Enter hostname or IP address: 8.8.8.8
Reply from 8.8.8.8 : bytes = 30 time = 0.66 ms
Reply from 8.8.8.8 : bytes = 30 time = 0.39 ms
Reply from 8.8.8.8 : bytes = 30 time = 0.35 ms
Reply from 8.8.8.8 : bytes = 30 time = 0.36 ms
```

Result :

Thus, a Ping program was successfully implemented using Python to send ICMP Echo Request packets, receive replies, and calculate the round-trip time (RTT) to check network connectivity.